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# Video-based automatic people counting for public transport: On-bus versus off-bus deployment☆

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ABSTRACT

Interest in Automatic People Counting (APC) for crowd detection and management is rapidly growing. While a range of Internet of Things (IoT) sensors and systems exist, video analytics is emerging as a particularly attractive option — especially for applications where more traditional methods of people counting are not available, unreliable or expensive. In this paper we focus on automatic people counting in the public transport context – specifically, rail replacement bus services – in which bus companies are typically contracted to provide bus services to replace train services during periods of planned and unplanned line disruption. This presents a particularly compelling use-case for video-based people counting, while also presenting unique challenges. Field trials are thus vital to the proper assessment of video-based APC solutions, however remain relatively scarce in the literature. While datasets to support research and benchmarking exist, these do not capture the intrinsic complexities of real-world deployment and the implications of selected configurations — in particular, on-vehicle versus off-vehicle use cases. In this paper, we evaluate our own video-based APC solution, representative of state-of-the-art approaches in the literature, in two separate extensive (i.e, multi-day) metropolitan field trials covering both on and off-bus use-cases. Through real-world deployment of the system in both settings, we highlight key differences with respect to APC accuracy, as well as other practical considerations, and the validity of underlying assumptions in both on and off-bus scenarios.

## 1. Introduction

Overcrowding and under-utilisation of vehicles are equally challenging problems for public transport systems globally (Sipetas et al., 2020). As a result, Automatic People Counting (APC) has emerged as an area of growing interest for the use of video analytics, as public transport providers seek more accurate, scalable and flexible solutions for measuring and predicting service demand and crowd management requirements (Komatsu et al., 2021; Liu et al., 2018; Nagaraj et al., 2022; Zheng et al., 2020). While numerous APC technologies exist, camera-based solutions leveraging advanced computer vision are emerging as a particularly attractive option due the relative low-cost, scalability and flexibility of vision as a sensor (Kuchár et al., 2023). In particular, the development of real-time video analytics algorithms to support the analysis of passenger counting and patronage analysis generally, deployable "on the edge" is rapidly growing in interest. However, the varying data requirements, the desired response speed, the environmental context of data capture, the quality (and variability) of video capture devices, among myriad other considerations, all impact the design and choice of video analytics algorithms for crowd and people movement analytics, and how they are integrated within larger Intelligent Transport Systems (ITS).

In this paper we consider video-based APC in the context of rail replacement bus services, in which bus companies are typically contracted to provide buses to replace trains during periods of rail line

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disruption, planned and unplanned. In this context, demand forecasting is essential to avoid both under-procurement and over-procurement of bus services, but also to plan for crowd management strategies at major passenger pick-up locations - all made difficult by a lack of reliable data and minimal, if any, real-time feedback on service demand and/or passenger wait times (Ji et al., 2015). Moreover, the temporary nature of rail replacement passenger pick-up/put-down zones, and variability of buses typically commissioned to provide rail replacement services demands APC solutions that may be quickly installed and setup, and then un-installed upon completion of the service. Video-based APC thus offers a scalable, flexible and versatile solution to meet these requirements (Velastin et al., 2020; McCarthy et al., 2021).

Video data is information-rich, but complex to analyse. Thus, sophisticated video processing algorithms are required, often leveraging the power of machine learning (and in particular deep neural networks), to reliably and robustly extract this information (Kumar and Raubal, 2021). Most commonly, such algorithms are bench-marked using pre-recorded datasets with human-annotated ground-truth people counts (e.g., Yan-yan et al., 2014; Perng et al., 2016; Liu et al., 2017). However, such datasets typically only portray a limited range of possible video-based APC configurations. For example, datasets for passenger counting overwhelmingly focus on on-vehicle use cases – i.e., cameras mounted inside the vehicle – and generally only consider over the doorway camera mounting positions. In practice, however, cameras can be mounted in a variety of locations, indoor and outdoor, as well as inside and outside of vehicle, sometimes pre-installed for other purposes such as on-bus security (e.g., Sutopo et al., 2021; Dahlan et al., 2021). Such configurations are associated with unique challenges and characteristics that may impact the validity of any underlying assumptions, and therefore overall system performance (Zhang et al., 2022). Off-vehicle and outdoors, for example, expose the system to lighting variations. On-vehicle, conversely, places the camera within the confined space of a moving vehicle, thus potentially limiting visual coverage. These practical differences have implications for the choice of video analytics method deployed, the hardware used, its installation and configuration, and indeed the suitability of video-based APC generally for the envisioned use case.

Of particular interest in the rail replacement bus context is on-bus versus off-bus passenger counting. That is, the choice to perform video-based APC using cameras installed on the vehicle, or using ground-based cameras at dedicated passenger pick-up/put-down zones. However, the two scenarios serve the purpose of measuring demand and embody different priorities. Off-bus counting focusses on demand at a specific location, prioritising the analysis of queuing passengers, and thus potentially other information such as passenger wait times and pedestrian flow more generally. In contrast, on-bus counting directly serves the purpose of maintaining ongoing real-time per-bus occupancy updates. Both scenarios also represent fundamentally different conditions in which video-based APC solutions can be deployed. Although video-based APC evaluations and associated people movement and social distance estimation methods are increasingly being reported (e.g., Hsu et al., 2020a; Dahlan et al., 2022; Mokayed et al., 2023; Marczyk et al., 2024), most consider only a single modality of use, most commonly in vehicle use cases, and with limited exploration of the impacts of other configuration and installation choices using the same method. To date, there has been no APC evaluation that explicitly examined the impact of deploying the same video-based APC system under both on- and off-vehicle modalities. We argue that this represents a fundamental configuration choice for video-based APC with potentially significant implications for the design and operating limits of such solutions. Moreover, the choice of video-based configuration has implications with respect to crowd management use-cases beyond just counting.

To this end, we compare video-based APC for both on-bus and off-bus use-cases, drawing on two extensive field trials; both undertaken over 7 days in two major Australian cities (Sydney and Melbourne). For on-bus use, we implemented and deployed a video-based APC solution on bus services within Sydney's metropolitan bus network. For off-bus, the same APC system was made weather-proof and deployed at several bus passenger pick-up locations throughout Melbourne's metropolitan public transport network. Both trials were carried out in the context of the implementation of planned rail replacement bus service and as part of larger projects that sought to compare different APC technology choices (McCarthy et al., 2021; Shahhoseini et al., 2022). In so doing, this paper makes the following novel contributions:

1. The first exploration and evaluation of key configuration choices for a low-cost, compute-efficient video-based passenger counter (i.e., on-bus versus off-bus installation, and camera position choices) in the context real-world passenger-facing bus services. To the best of our knowledge, this is the first such field trial to specifically compare the performance of an APC solution for the same principal task (passenger counting), but under fundamentally different modalities of use.
2. Novel insights into the practical and operational factors that influence and impact the accuracy and performance of line-of-interest passenger counting using video-based APC methods. Through this, we offer practical guidance for the implementation and evaluation of video-based APC solutions and assessing their suitability for specific configurations and use-cases.

The paper is structured as follows. Section 2 reviews the existing literature on APC, focusing specifically on video-based systems, and previous field trials. Section 3 presents the details of our video-based APC method, and the two field trials in which it was deployed for on-bus and off-bus (ie., on-ground) applications, respectively. Section 4 presents results from both trials, and compares the performance of the system with respect to accuracy, as well as other key performance indicators. Section 5 discusses the broader implications of the two deployment scenarios with respect to operational, business and managerial considerations. Finally, Section 6 states our conclusions and directions for future research.

## 2. Related work

### 2.1. Automatic people counting and monitoring

Significant research as been undertaken in the development and evaluation of real-time Automatic People Counting (APC) solutions, each exhibiting various trade-offs and being applicable under certain conditions. Radio frequency-based methods, for example, obtain counts via fluctuations in power level or characteristics of the signal propagation from transmitter to receiver, or back-scatter of the signal due to the presence of people (Kouyoumdjieva et al., 2019). Such methods offer varying levels of accuracy, but require additional infrastructure to support. Various sensor-based solutions also exist, such as Passive Infrared (PIR), in which people counting is acquired from temperature changes (e.g., Wahl et al., 2012; Xie et al., 2023), or light-emitting diodes (LEDs) in which a visible light beam is interrupted by objects present (e.g., Yang et al., 2017). Wi-Fi based methods such as Delzanno et al. (2023), Myrvoll et al. (2017) and Jiang et al. (2023) are another set of solutions exploiting the presence of Wi-Fi hotspots to analyse probe-request frames as a means of determining counts. This however assumes the presence of single devices on each person, the presence of a Wi-Fi hotspot (or the setting up of one), and typically requires variable time periods to acquire an accurate count.

### 2.2. Video-based APC

Video data offers an unparallelled bandwidth of information and versatility of use-cases for people counting and monitoring applications. In contrast, most non-video-based APC methods described above offer only a single modality of data, with limited ability to support

richer analysis of people movement. While exceptions such as LiDAR exist (Lesani et al., 2020), the need for specialised and most often expensive sensing and/or supporting infrastructure makes such solutions difficult to deploy at scale (Kouyoumdjieva et al., 2019). Such considerations, in combination with the relative low cost and/or ubiquitous presence of cameras in public spaces (or easy installation thereof), has motivated the development of video analytics solutions for both APC and urban analytics use-cases more broadly. However, unlike most other sensor modalities, video data requires substantially more processing to extract and utilise the visual information present.

Video-based passenger counting represents a specific use-case of the more general people/crowd counting and monitoring problem in which approaches can be categorised as either: line-of-interest "enter/exit" (or passenger flow) counting, and region-of-interest "head counting" (Labit-Bonis et al., 2021). Obtaining enter-exit counts thus amounts to a detection and tracking problem, in which people counting seeks to determine the net occupancy of an area (typically a vehicle such as a bus in the passenger analytics context) based on monitoring of passenger movement. As such, the video analytics solution is focused on the robust determination of each person's movement within the camera's view. Such methods are typically referred to as *detection-based* methods (Labit-Bonis et al., 2021). Conversely, region of interest (ROI) head count methods (i.e., crowd counting and density estimation methods, (e.g., Fang et al., 2019; Liu et al., 2018; Khan et al., 2023; Tian et al., 2019; see Gao et al., 2020 for a recent review) seek to determine the total number/density of people occupying a space at a given time instance, most commonly via regression-based methods.

In the context of real-time, low-cost video-based APC, however, detection-based methods remain the dominant approach, and thus, are the focus of this work.

### 2.3. Detection-based APC methods

Early detection-based methods typically first extracted hand-crafted features and/or performed background subtraction to determine the presence and location of either whole bodies, or heads within video frames. For example, Histogram of Gradients (HOG) (Dalal and Triggs, 2005) has had a long association with people detection and has seen wide range of ITS applications more generally (e.g., Gandhi and Trivedi, 2007; Velastin, 2009), offering a local shape descriptor based on edge orientations. Numerous other patch-wise features have been proposed, reflecting the focus of the period on identifying new and effective ways to robustly capture object features to further boost classification performance. More recently, Yan-yan et al. (2014) propose using HAAR features to detect people, and demonstrate its effectiveness on pre-recorded video feeds from different camera views.

In recent years, video-based APC has been dominated by deep learning methods for person detection, tracking and re-identification - in particular, the use of pre-trained Convolutional Neural Networks (CNNs) (Sindagi and Patel, 2018). Common pre-trained models applied for video-based APC include Faster R-CNN (Ren et al., 2017), MobileNet SSD (Howard et al., 2019), and variants of the "You Only Look Once" YOLO model (Redmon and Farhadi, 2018; Bochkovskiy et al., 2020; Zhao et al., 2022; Tian et al., 2022). Such models are often paired with tracking and re-identification methods. For example DeepSORT (Wojke et al., 2017), an algorithm for multi-object tracking in video streams, has proven to be very effective for multi-person tracking in the presence of occlussions, and with more recent further improvements to this approach (Du et al., 2023). Re-identification models are also typically employed to support accurate tracking of individuals in line-of-interesting counting (e.g., Zhao et al., 2022; Sutopo et al., 2021).

However, the choice of models and methods to support detection and tracking typically forces a trade-off between accuracy and speed of execution, which in turn is dependent on the extent of robustness to variability in environmental conditions (weather, lighting, occlusions, crowd density), as well as camera angle, and the underlying hardware to support its use on the edge. For example, Komatsu et al. (2021) apply the Faster R-CNN network, a model known to achieve high accuracy for people detection. However, the method is only applicable to off-line analysis.

### 2.4. Video-based APC evaluations and field trials

Off-vehicle APC studies most often feature railway stations and/or transit locations. In many cases, pre-existing video surveillance cameras are utilised, motivated by the low cost and value-adding opportunity such systems could provide. For example, Yan-yan et al. (2014) consider the application of detection-based counting using fixed-in-place video surveillance footage from a metropolitan railway station (Xizhimen, China). They propose a more generalised approach using HAAR features and demonstrate its effectiveness on pre-recorded video feeds, noting the challenge of performing robust video-based counting from non-optimal and variable camera angles. Dahlan et al. (2021) consider the use of CCTV cameras for video-based people monitoring within a railways station in the context of social distancing in response to COVID-19.

More recent field trials have focused increasingly on the use of fixed-in-place installed-for-purpose cameras for ground-based counting. Zheng et al. (2021), for example, present a dataset of videos captured from cameras mounted above a Platform Screen Door (PSD), selected as an optimal view to avoid occlusions. They train and evaluate a detection-based CNN method using the dataset. To address the lack of representative datasets upon which to train and evaluate video-based APC solutions, Velastin et al. (2020) present PAMELA, a multi-sensor facility designed explicitly to assess video-based people monitoring for different application domains, including public transport. The facility supports controlled assessment of methods under different lighting conditions, including simulated daylight, and moonlight. A dataset is also presented, simulating a London underground station, with annotated camera-footage obtained from cameras depicting various angles. The authors demonstrate that such variations of view can cause substantial loss of accuracy for video-based APC methods using pre-trained models (eg., YOLOv3 Redmon and Farhadi, 2018).

In the on-vehicle context, most attention has been given to bus passenger counting (Hsu et al., 2020a). Perng et al. (2016) propose and trial a single camera detection-based counter on a metro bus, examining the impacts on the accuracy of APC under different lighting conditions, with the accuracy of the passenger count varying from 75% to 90%. (Liu et al., 2017) evaluate the performance of a detection-based CNN for video-based APC on pre-recorded surveillance video captured from metropolitan bus services. Videos depict scenes from camera locations above front and rear bus doors, with a focus on how the proposed method handles tracking under high passenger density in the doorway area. Hsu et al. (2020b) compare numerous detection-based APC methods in the on-bus context, with specific attention given to the impact of external environmental conditions (that is, day, night, and rain). The results indicate variations in accuracy for most methods under different conditions. (Haq et al., 2020) field trial an embedded solution based on HOG feature extraction and SVM-based classifier, and KLT tracker. The solution is installed above the doorway of a single bus, achieving an accuracy of 94.5%. Focussing only on low-powered edge computation solutions for people detection, Ghaziamin et al. (2024) report comparison results achieved using various detection models, including YOLOv4 and Faster-RCNN over datasets created from indoor overhead cameras. While they do not report count accuracies, they report a mean Average Precision (mAP) of 87% for the best performing YOLOv4 model. Cheng et al. (2014) evaluate a trajectory-based method for line-of-interest (bus entry/exit) counting in the on-bus context. An hour of footage captured from a bus rear door is used for evaluation; however, only bus exits occurred through this doorway. The authors

note that the evaluation footage contains significant light change and, at times, low illumination.

Previous work has compared and evaluated four different APC technologies, including video-based field trials in buses (McCarthy et al., 2021). In this extensive 1 week study, each technology (infrared, Wi-Fi, a pressure sensing mat, and a video-based APC solution) was installed on metro bus services. The video-based APC was shown to perform well compared to infrared and WiFi-based counting and to be comparable to floor-based pressure sensing with respect to per-stop enter/exit count accuracy. However, the study considers only the on-bus scenario, and does not provide an in-depth analysis of the impact of different camera configurations.

In summary, while video-based APC has been well studied with respect to specific use cases, there currently exists no study that compares the performance of video-based passenger counting in different configurations (i.e., on-vehicle versus off-vehicle), nor extensively evaluated the impact of different camera positions. Previous studies clearly show the unique and distinct challenges of both modalities, which warrants such an evaluation.

## 3. Methodology

Two field trials were independently conducted in which the same video-based Automatic People Counting (APC) method was deployed for on- and off-bus people counting. Both trials were undertaken as part of larger projects seeking to evaluate and compare the accuracy of IoT technologies for passenger counting (the on-bus project has been previously reported in McCarthy et al., 2021). Here, we focus specifically and in more depth on the performance of a video-based solution. Specifically, we implemented a detection-based person counter in both on- and off-bus passenger counting scenarios. We describe the video-based APC solution below, and then outline the two field trials in which the system was deployed for on- and off-bus passenger counting, respectively.

### 3.1. Video-based APC

The APC solution deployed consisted of a video-based automatic people-counting algorithm designed for use with standard RGB video frames and taking advantage of a pre-trained *Convolutional Neural Network* (CNN) for people detection and tracking. Although there are numerous methods for performing image / video-based people counting, a detection-based 'line of interest' counter was implemented based on its suitability for real-time use on low-cost, resource-limited hardware, and as a method representative of the latest for detection-based counting, comprising a standard pipeline of detection → tracking → reidentification) (Gawande et al., 2020). From a practical perspective, detection-based methods have previously been demonstrated to offer versatility of uses with respect to camera positioning and image quality, and ease of calibration, making the approach a suitable choice for the different configurations explored in the two field trials reported here. Thus, while more advanced methods exist, the method implemented was selected as a generally applicable method, and one that may be readily integrated with existing camera systems.

Fig. 1 provides an overview of the video-based APC system's processing pipeline. The major steps of the algorithm are described below.

#### 3.1.1. Person detection

Person detection is achieved using the well-established edge compute-targeted pre-trained lightweight neural network, MobileNetv3 Single Shot Detector (SSD) (Howard et al., 2019). Specifically, MobileNetv3 is applied to each incoming frame and is used to generate bounding-box predictions of people instances in each consecutive frame. A positive person detection is generated for bounding box predictions with an attributed confidence level greater than a set threshold, which was set to 50% for both trials.

#### 3.1.2. People tracking

As is typical for detection-based methods, predicted bounding boxes from the person detector form the basis of tracking. For this, the bounding box of each new detection is compared with the bounding boxes of the track objects of the previous frame. New detections that are deemed to have a highest affinity based on appearance and location (discussed further below) are assumed to be a match. A positive match triggers an update of the track object's position using the new detection's location, with the associated confidence of the track object set to the maximum of the current detection confidence and that previously registered with the track object.

Where no match is found, then a new track object is created and added to the list of objects to track, with the new track object's location and confidence set to that of the positive detection. Track objects with no new detection registered are reduced in confidence by 10% confidence each update, and deleted from the list when confidence falls below 20%.

#### 3.1.3. Track object affinity

A critical component of the tracking phase is accurately matching new detection instances with previous track objects. This was hypothesised to be particularly important for the expected close proximity of people at bus entry/exit points, where detection-based methods have been observed to be susceptible to incorrectly attributing a new detection to a different (possibly occluding) track object in close proximity, causing track objects to swap, and ultimately incorrect counts to be generated. To address this, our implementation employs two metrics based on distance and appearance to determine the correct matching. Specifically:

1. *Intersection over Union* (IoU) with a previous track object emphasising the assumption of small track object movement between frames, and
2. the cosine difference of feature vectors generated from both bounding-box regions using OpenVino's pre-trained re-identification network: `person-reidentification-retail-0031`, based on Intel's RMNet backbone network (Izutov, 2018). This model was selected on the basis of being lightweight and its proven ability to support accurate re-identification in real-time, resource-limited settings.

A final affinity score is generated from the multiplication of both scores, and a match is deemed positive if it exceeds a set threshold (set to 0.5 for both trials).

#### 3.1.4. Counting

The final step of the algorithm is the counting, defined over a *region of interest* (ROI), where the counts of 'IN' or 'OUT' are increased based on the direction of each tracked person and their complete traversal of the ROI. A track object that passes through the ROI and back again is not counted.

Four virtual lines are defined to form a quadrilateral ROI in image space (with the option of using only two virtual lines and the image boundaries). When a track object passes from one side of the ROI to the other (i.e., crosses both opposing virtual lines), the appropriate in/out count is incremented. A gap between the lines is used (and calibrated for each scenario) to ensure robustness to hysteresis from tracking, and in general as an additional filter against false-positive detections.

### 3.2. Field trial configurations

To evaluate and compare the performance of the video-based APC method for passenger counting on- and off-bus, the results obtained from two separately conducted field trials using the same method were pooled and analysed. For on-bus passenger counting (*Trial 1*), the video-based method was used to count bus passenger entry and exits on two metropolitan buses that service routes within Sydney, Australia.

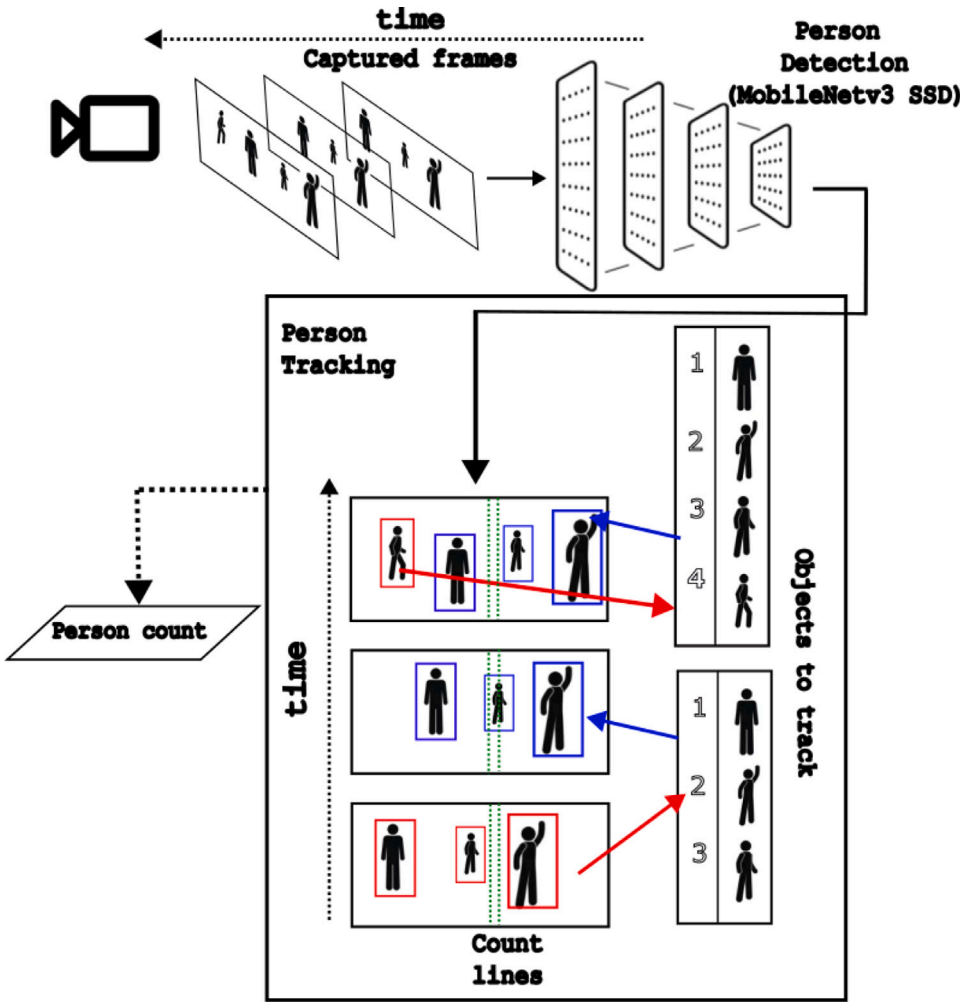


**Fig. 1.** The video-based Automatic People Counter processing pipeline. Video frames are captured and fed through a pre-trained CNN (MobileNetv3 SSD), producing person detections in each frame. Detections above a given confidence threshold are compared with current list of tracked persons to determine if the detection is new (red bounding box), in which case it is added to the list, or an existing tracked person (blue). Tracking determines when each person crosses a calibrated count of line of interest (dotted green line), from which "In" and "out" counts are determined. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

For off-bus passenger counting (*Trial 2*), the same method was used to count bus passenger entries at a Rail Replacement Bus pickup zone at various locations within metropolitan Melbourne, Australia.

For Trial 1 (on-bus), two cameras were installed inside each bus, with each camera facing outward in the direction of the front and rear doors, respectively. These camera positions were determined to provide sufficient visual coverage of the front and rear entry/exit points, respectively. Fig. 2 shows the device installed in each bus and an example location of the device's installation.

For Trial 2 (off-bus) passenger counting, two cameras were installed on tripods for the primary purpose of counting passengers boarding bus services during planned rail disruptions. Two cameras were used to explore different viewing angles with respect to the queue of passengers boarding the bus. Fig. 4 shows example images of the technology in use during bus trials.

While the video-based APC solution is capable of real-time processing on streaming video data, a post-processing methodology was adopted for both trials. This approach was chosen for two reasons: 1. to reduce engineering requirements and the risk of data loss due to software/hardware failure; and 2. To facilitate evaluation against ground truth obtained from the recorded video data. We note, however, that all processing of the recorded video was performed in real time, and under the same computational constraints as would be available at the point and time of capture using our device. For Trial 1, the camera feeds were captured at a resolution of 640 × 480 pixels, and a frame rate of 30 frames per second, with post-processing performed on a *Raspberry Pi 3B+ single-board computer* (SBC), featuring a Movidius NeuroCompute GPU stick, powered by a standard LiPo Powerbank. For Trial 2, a *Raspberry Pi 4B* SBC was used for data capture, again powered by a LiPo Powerbank, and connected to a SBC camera module v2 acquiring images at 1920 × 1080 pixel resolution at 25 frames per second. For both trials, video data was saved to an SD card mounted on the SBC.

Institutional ethics clearance was granted by the Human Research Ethics Committee of Swinburne University of Technology for both the on-bus trial (2018/302) and the off-bus trial (20213060-6326). Further details of both trials are provided in the next section.

## 4. Results

In this section we present results from both Trial 1 (on-bus) and Trial 2 (off-bus). We first provide the details of each trial, and the data collected from both trials before presenting an analysis of the video-based APC performance in both settings. We present the analysis in two parts. First, we evaluate and compare the performance of the system with respect to the accuracy of total passenger counting. We then present use-case specific quartile accuracy analyses, focussing on the ability of the video-based APC to provide ongoing estimates of bus occupancy based on passenger exchange estimates per stop and per

**Fig. 2.** Video-based APC device used for on-bus trials (left) and its installed location for front door monitoring (right), circled in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

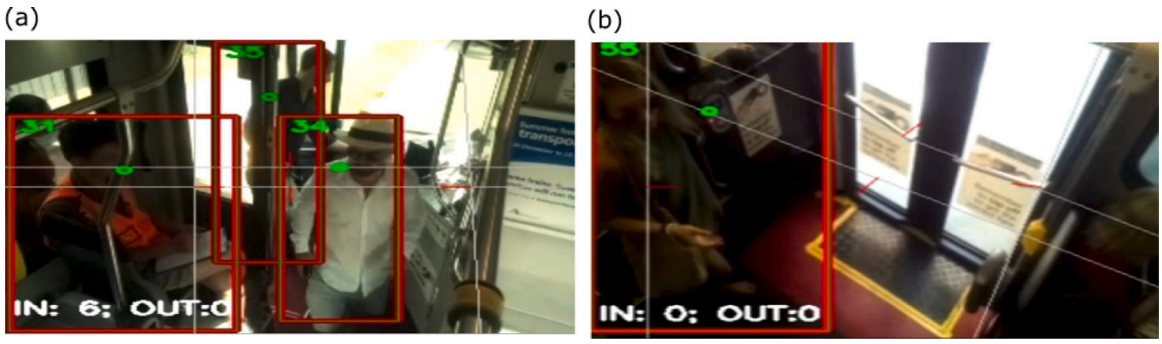


**Fig. 3.** Example processed frames from the video-based counter during the on-bus Trial 1 showing (a) the front door virtual count line (light grey horizontal section within central region of image), and (b), the rear door virtual count line (light grey angled line from top left to bottom right of the central image region). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

bus, for the on-bus and off-bus use cases, respectively. Through this we seek to evaluate the effectiveness of the video-based APC to fulfil these distinct data capture needs, and to understand the impact of deploying the same video-based APC in these fundamentally different settings.

### 4.1. Trial 1 - on-bus passenger counting field trial (Sydney)

For Trial 1, the video analytics solution was trialled over 7 days in Sydney, Australia (December, 2018). At the start of each day, two cameras were installed inside a bus, each camera facing outward in the direction of the front and rear doors respectively. Fig. 2 shows the device installed in the bus and example location of the device's installation. From Monday to Friday, data collection took place from both cameras over approximately 8 h each day, between 7.30 am and 3.30 pm, as the bus serviced standard metropolitan routes around the inner South Eastern suburbs of Sydney. On the weekend, the technology was trialled on rail replacement bus services between Sydney's Central Station and Bondi Junction railway stations. Installations took place before the buses departed the local Bus depot, and were uninstalled upon returning at the end of the day. The installed devices in the bus were accompanied and monitored by a research team member at all times (see Fig. 3).

### 4.2. Trial 2 - off-bus passenger counting field trial (Melbourne)

The off-bus trial was carried out over two separate rail disruptions, each involving separate pick-up locations in the morning and afternoon, in metropolitan Melbourne. Data capture during Disruption 1 was conducted over 4 days between 6 May and 11 May 2021 (weekdays only), with 8 h of data capture per day at two separate locations (Reservoir and Parliament Stations in Melbourne's metropolitan rail network). Data capture during Disruption 2 was conducted similarly over 3 days between March 9th and March 15, 2022, at Pakenham and Dandenong Stations. Both data capture events involved installing technology options at Express Rail Replacement Bus pickup zones for the primary purpose of counting passengers boarding bus services during these planned disruptions. Fig. 4 shows example images of the technology in use during the off-bus trials.

Two cameras were used throughout the trial, both mounted on tripods. *Camera 1* was always positioned close to the road-side, pointing in a direction parallel to the side of the bus when it stopped at the passenger pick up. *Camera 2* was introduced in the first two days of trial to offer an alternative location for counting passengers, positioned further away from the road-side, and angled towards the passenger queue. On later days, however, this camera was removed from passenger counting duties. Fig. 5 shows sample processed frames from both cameras, at different Trial 2 sites.

### 4.3. Data collection and analysis

#### 4.3.1. Ground truth

The *actual per-bus passenger occupancy*, or ground truth ($n_{gt}$), was collected by personnel during data capture, and for selected periods, subsequently validated by video inspection. For Trial 1 (on-bus), ground truth was obtained by annotating the time the bus arrived at each bus stop, and then separately counting how many passengers entered and alighted through either door. For Trial 2, ground truth counts were obtained exclusively by inspecting the captured video footage. Bus arrivals and departures were recorded at each passenger pick-up location and the total number of passengers who boarded each bus was also recorded. As the focus for Trial 2 was only on the boarding of passengers through the front door of the bus (the rear door remained closed), ground-truth counts were only obtained for front-door entries for the off-bus trial.

**Fig. 4.** The off-bus video-based counter device (left) and the "Camera 1" positioning used throughout Trial 2 (right).

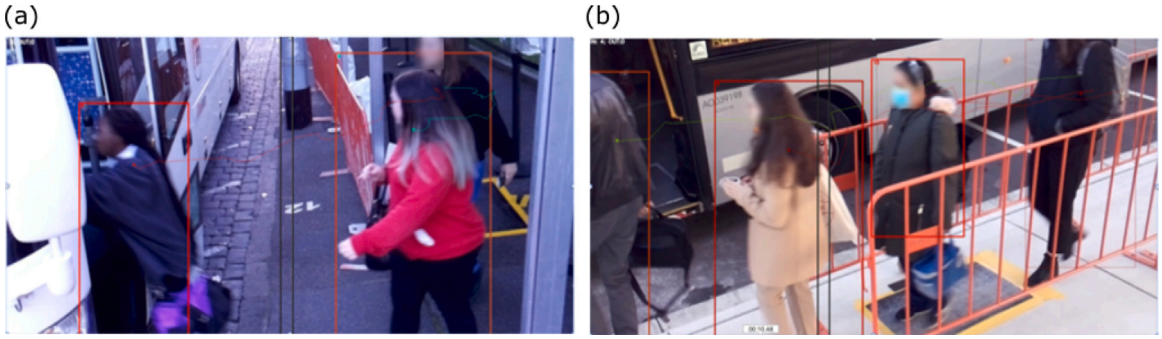


**Fig. 5.** Example processed frames from the video-based counter during the off-bus Trial 2 showing (a) Camera 1 facing side on to the boarding passenger flow, and (b) Camera 2: at an approximately 45 degree angle to the boarding passenger flow. Images in both rows show samples from different trial site locations from that camera perspective.

After completion of data collection, the data collected was inspected to identify viable data for analysis, and then aligned collected data with the corresponding ground truth data collected during the trial period.

#### 4.3.2. Accuracy metrics

The primary metric of interest is the accuracy of the *net passenger counts* ($\hat{n}$) over a given data capture period. The accuracy can be computed over different segments of the data: per camera, per location, etc., and was calculated as follows:

$$\text{Accuracy} = \left( \begin{cases} 1 - \frac{|n_{gt} - \hat{n}|}{n_{gt}}, & n_{gt} \geq \hat{n} \\ 1 - \frac{|\hat{n} - n_{gt}|}{\hat{n}}, & n_{gt} < \hat{n} \end{cases} \right) \times 100\%, \tag{1}$$

where $n_{gt}$ is the ground truth number of passengers, and $\hat{n}$ is the estimated number.

In addition, standard error measures were also calculated, including the Mean Absolute Error (MAE):

$$\text{MAE} = \frac{\sum_{i=1}^{N} |n_{gt} - \hat{n}|}{N}, \tag{2}$$

and the Root Mean Square Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N} (n_{gt} - \hat{n})^2}{N}}, \tag{3}$$

where $N$ is the number of recorded passenger exchanges.

In addition to net passenger counts, each trial also performed a finer grained analysis of ongoing bus occupancy estimations.

For the on-bus trials (Trial 1), accuracy scores were calculated per stop, and are provided with mean and quartile values representing: (i) overall net passenger exchanges per stop (i.e., the accuracy of the method at estimating the number that entered the bus minus the number that exited the bus at each stop, across all data collected from both buses), with quartile accuracies for the range of net passenger exchange accuracies over all data; (ii) net passenger exchange accuracies per stop in finer detail, reporting accuracies for each door, and each passenger exchange direction (enter/exit) in isolation to see how the method performs at different aspects of the counting task, with mean and quartile accuracies given; (iii) total passenger count accuracies for each specific door and direction.

For off-bus trials (Trial 2), accuracy scores were also calculated and provided with mean and quartile values for occupancy estimates per bus, and thus an accuracy score was calculated for each bus that arrived during the trial, upon which an analysis of the quartile distribution was carried out. Note that passenger counts for off-bus trials account for passengers boarding the bus only. The above-mentioned metrics, together, provide a holistic view of the video-based APC system accuracy for both trial configurations.

#### 4.3.3. Calibration

The video-based people counter requires calibration of the Line of Interest (LOI) for counting, defined as a 4-sides region within the camera's view. The LOI for counting was configured separately for each session of data capture due to the need to re-install the counter prior to each session in both trials. In the on-bus setting, little change of calibration was required between sessions, however calibration was essential in the off-bus setting to account for day-to-day changes to the physical setup at passenger pick up locations. Once configured, the LOI was applied without modification across the entire video captured for that session.

#### 4.3.4. Data collection summary

Over the course of the two trials, a total of 152 h of camera footage was recorded. Video capture occurred continuously, with video files written every 5 min to the SD card of each accompanying Raspberry Pi. The recording between multiple cameras was synchronised so that the video files were time-stamped and corresponded to each other. A summary of the distribution of these hours across the two trials and at

**Table 1**
Summary of data collected for each day of both on and off-bus trials, including total time of data capture, number of passengers counts (PAX), number of passenger exchange events (N) and average number of passengers involved in each passenger exchange event (PAX/N). Note that in Trial 1 (on-bus), data was captured from cameras at the front and rear of bus however the time duration reported is time both cameras captured data simultaneously.

| Trial | Date | Location | Duration | PAX | N | PAX/N |
|---|---|---|---|---|---|---|
| Trial 1 | 12/12/2018 | Bus 2031 | 8 h, 45 min | 785 | 312 | 3.85 |
| | 13/12/2018 | Bus 2031 | 7 h, 45 min | 856 | 218 | 3.98 |
| | 14/12/2018 | Bus 2031 | 8 h, 25 min | 798 | 231 | 3.46 |
| | 15/12/2018 | Bus 2031 | 9 h, 21 min | 316 | 15 | 21.1 |
| | 16/12/2018 | Bus 2031 | 4 h, 45 min | 371 | 28 | 13.25 |
| Trial 2 | 06/05/2021 | Reservoir<br>Parliament | 3 h, 45 min<br>3 h, 55 min | 2,805 | 136 | 22.39 |
| | 07/05/2021 | Reservoir<br>Parliament | 3 h, 40 min<br>3 h, 50 min | 2,860 | 133 | 22.13 |
| | 10/05/2021 | Reservoir<br>Parliament | 2 h, 10 min<br>3 h, 55 min | 4,022 | 142 | 17.85 |
| | 11/05/2021 | Parliament | 3 h, 15 min | 1,720 | 73 | 16.16 |
| | 10/03/2022 | Pakenham<br>Dandenong | 4 h, 00 min<br>3 h, 30 min | 707 | 43 | 16.02 |
| | 11/03/2022 | Pakenham<br>Dandenong | 4 h, 00 min<br>4 h, 00 min | 660 | 49 | 14.81 |
| | 15/03/2022 | Pakenham<br>Dandenong | 4 h, 00 min<br>4 h, 00 min | 560 | 48 | 16.75 |

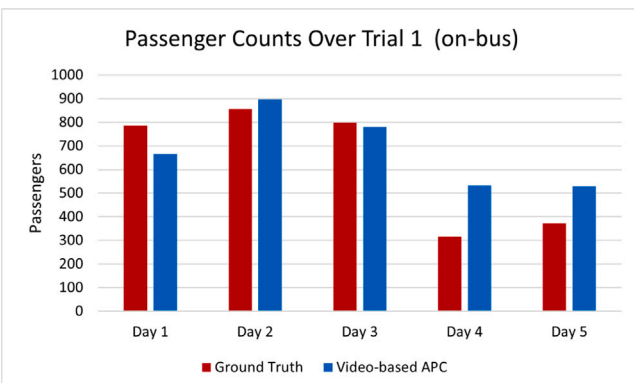


**Fig. 6.** Total passengers per day of Trial 1 that entered and/or exited the bus. Graph compares ground truth and the estimated count from the video-based APC algorithm. See Table 3 for exact passenger counts.

various locations is provided in Table 1. For the analysis presented here, the results of 95 h of the total data captured was used: 39 h from Trial 1 (on-bus), and 56 h from Trial 2 (off bus). These hours were selected on the basis of the quality of the captured data (i.e., no significant loss of data during capture periods due to hardware failure or other external factors), as well as the existence of validated ground-truth data (see Fig. 6). As noted, data collection was interrupted for periods in both trials due to a range of unforeseen causes, including hardware failure, weather-induced events, and field-trial maintenance needs. For completeness, Table 2 summarises these incidents and approximate duration of time impacted.

### *4.4. Passenger boarding/alighting event detection accuracy*

We now assess the overall accuracy of passenger counting in both contexts and with a focus on comparing the performance of the video-based APC system in both settings. Specifically, we compare the accuracy achieved with respect to the total passenger boarding/alighting events in both trials, as well as the MAE and RMSE error metrics, computed from the set of all passenger exchanges in both trials. We then report these metrics with respect to different data collection locations (that is, front and rear doors of the bus for Trial 1 and different off-bus passenger pick-up locations for Trial 2).

Tables 4 and 5 present passenger count precision, MAE and RMSE results for Trial 1 (on-bus) and Trial 2 (off-bus), respectively. In both tables, the final row reports the metrics achieved over all data collected in both trials, respectively. As can be seen, the overall accuracy achieved in Trial 1 (on bus) is 91. 8%. That is, the APC system estimated that 3,406 estimated passenger boarding or alighting events had occurred compared to the ground truth total of 3,126 passenger exchanges. Thus, over the 5 days of Trial 1 data collection, the on-bus video-based APC system over-estimated passenger boarding/alighting events by 280 passengers (8.2%). In contrast, off-bus trial results in Table 5 show a higher overall total passenger count accuracy of 96.6% is achieved, with the system correctly detecting 10,428 of the total 10,792 passenger boarding events (i.e., an under estimation of 3.3%).

In contrast to Accuracy, MAE and RMSE results reported in Tables 4 and 5 show higher errors recorded for off-bus counting (overall MAE: 4.91; RMSE: 7.66), compared with on-bus trials (overall MAE: 2.65; RMSE: 4.75). However, inspection of average passenger exchanges as reported in Table 1 makes clear that for Trial 2 (off-bus), substantially more passengers were involved in each bus boarding event compared with Trial 1's bus-stop-based counts. Thus, MAE and RMSE scores are unsurprisingly larger for Trial 2 exchange events; however, in general are of smaller proportion with respect to average passenger exchanges compared to Trial 1.

The above results make clear that the video-based APC performed differently in Trial 1 and Trial 2 with respect to net passenger count accuracy. However, it should be noted that the Trial 2 (off-bus) setting exclusively considered only boarding passengers and through only one bus entry point (a choice dictated by the associated partner's requirements). As such, the Trial 1 (on-bus) configuration represents an arguably more complex scenario for APC applications, with both boarding and alighting events in play, and through different entry/exit points (front and rear doors). Thus, it is of interest to consider the breakdown of accuracy scores from Trial 1. To this end, Table 4 provides a breakdown of the accuracy scores for each doorway and each event type of Trial 1. The highest accuracy achieved is 97.6%, corresponding to counts of boarding passengers through the front door. Notably, this accuracy is almost identical to the Trial 2 system's reported counting accuracy for off-bus passenger boarding. In contrast, the on-bus system's accuracy for passenger exiting events through the

**Table 2**
Summary of incidents causing data loss or data quality reduction over both trials.

| Trial | Date | Location | Duration | Cause |
|---|---|---|---|---|
| Trial 1 | 10/12/2018 | Bus 2031 | All Day | Trial Camera configuration and testing |
| | 11/12/2018 | Bus 2031 | All Day | Trial Camera configuration and testing |
| | 12/12/2018 | Bus 2031 | 50 min | No power - Front Door Cam |
| Trial 2 | 06/05/2021 | Reservoir<br>Parliament | 10 min<br>15 min | Camera adjustments & intermittent recording failure |
| | 07/05/2021 | Reservoir<br>Parliament | 10 min<br>10 min | Camera adjustments & intermittent recording failure |
| | 10/05/2021 | Reservoir<br>Parliament | 1 h<br>5 min | SD Card failure (Reservoir) & camera adjustments (Parliament) |
| | 11/05/2021 | Reservoir<br>Parliament | 20 min<br>5 min | Heavy rain induced camera adjustments (Reservoir), intermittent recording failure (both) |
| | 10/03/2022 | Pakenham<br>Dandenong | 1 h<br>30 min | Insufficient light (pre-sunrise Pakenham), camera tripod stability due to wind (Dandenong) & camera adjustments (Parliament) |
| | 11/03/2022 | Pakenham<br>Dandenong | 1 h<br>0 min | Insufficient light (pre-sunrise Pakenham) |
| | 11/03/2022 | Pakenham<br>Dandenong | 1 h<br>0 min | Insufficient light (pre-sunrise Pakenham) |

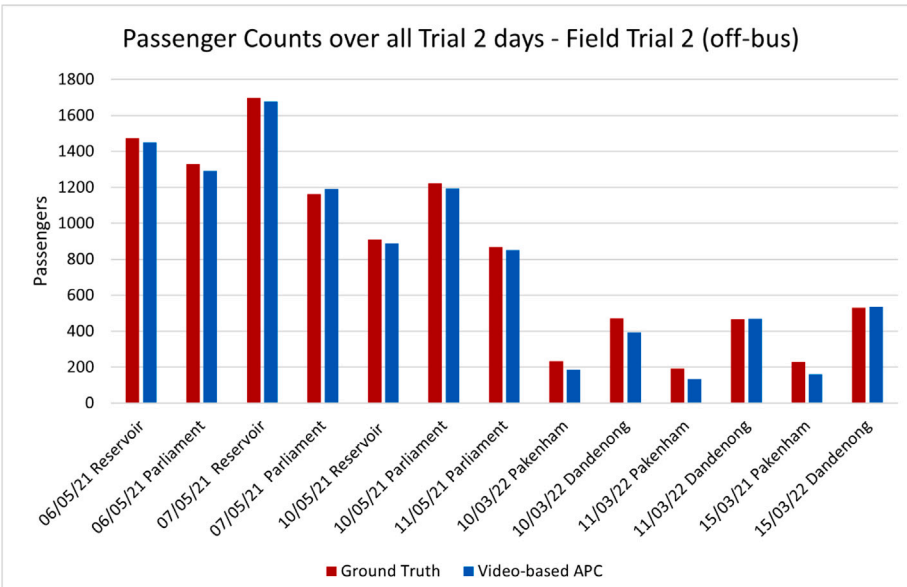


**Fig. 7.** Total passengers who boarded a bus for each day of Trial 2 (off-bus). Graph compares ground truth and the estimated count from the video-based APC algorithm. See Table 3 for exact passenger counts.

same door drops to 88.4% accuracy, an approximately 10% reduction in accuracy, suggesting a clear bias towards passenger boarding events in the on-bus context.

Trial 1 "Rear In" and "Rear Out" results suggest the rear door counting was the most challenging for the video-based APC. While "Rear Out" events were the only authorised direction of passenger flow on most bus services (that is, passengers were in general not allowed to enter the bus via the rear door), the bus system estimated a total count of 318 passengers entering the bus via the rear door, compared to a ground truth of only 39 (that is, a 12.26% accuracy). Notably, "Rear Out" event detections yielded the highest net passenger count accuracy of 99.92%. However, MAE and RMSE results offer a different perspective, with "Rear Out" estimates achieving a relatively high MAE and RMSE, indicating that while the net count was close to ground truth, per-stop passenger counts through the rear door were, in general, not as accurate as the front door scenarios, though clearly centred on a reasonably accurate mean suggesting a balance of under and over counting was occurring. In contrast to this, "Front In" achieved a similarly high accuracy, but also a much lower MAE and RMSE, suggesting that overall the video-based APC was performing best for counting passenger entries through the front door. Notably. MAE and RMSE scores for on-bus front door exits are lower than for front door entries; however, this is likely due to the much lower numbers of passengers who exited (412) compared to entered (1539) through the front door, suggesting the "Front Out" count was most frequently 0. Further analyses of front-door versus rear-door counts are provided in Section 4.5 below.

The accuracies of the passenger boarding counts in Trial 2 are also broken down across each day of the trial in Table 3 (see also Fig. 7), and across the four data collection locations in Table 5. In general, off-bus passenger boarding accuracies remain stable across the four locations, with the exception of Pakenham where a substantial 25% reduction in accuracy is made evident. Inspection of Table 2's summary of data capture loss and reduced data quality makes clear this venue experienced significant periods of low light (1 h of each of the 4 h data capture sessions). This and other use-case specific aspects of the off-bus system are further analysed in Section 4.5

**Table 3**
Trial 2 (off-bus) daily passenger counts, showing average ground truth passenger counts ($n_{gt}$), APC estimated ($\hat{n}$) passenger counts, and associated accuracy, MAE and RMSE scores for each location. See Fig. 7 for a graphical representation of this data.

| Day | Date | Location | $n_{gt}$ | $\hat{n}$ | Accuracy (%) | MAE | RMSE |
|---|---|---|---|---|---|---|---|
| 1 | 06/05/2021 | Reservoir | 1475 | 1450 | 98.3 | 5.37 | 10.50 |
| | | Parliament | 1330 | 1292 | 97.14 | 5.71 | 15.50 |
| 2 | 07/05/2021 | Reservoir | 1698 | 1678 | 98.82 | 4.42 | 7.48 |
| | | Parliament | 1162 | 1193 | 97.33 | 4.32 | 7.10 |
| 3 | 10/05/2021 | Reservoir | 909 | 888 | 97.69 | 7.89 | 14.53 |
| | | Parliament | 1223 | 1195 | 97.71 | 3.36 | 5.01 |
| 4 | 11/05/2021 | Reservoir | – | – | – | – | |
| | | Parliament | 868 | 852 | 98.16 | 7.40 | 10.64 |
| 5 | 10/03/2022 | Pakenham | 234 | 184 | 78.63 | 4.74 | 7.17 |
| | | Dandenong | 473 | 395 | 83.51 | 3.83 | 6.00 |
| 6 | 11/03/2022 | Pakenham | 193 | 135 | 69.95 | 3.39 | 6.68 |
| | | Dandenong | 467 | 470 | 99.36 | 2.41 | 3.69 |
| 7 | 15/03/2022 | Pakenham | 230 | 160 | 69.57 | 4.36 | 6.47 |
| | | Dandenong | 530 | 536 | 98.88 | 2.17 | 6.47 |

**Table 4**
On-bus passenger count accuracy as percentage of total passenger exchanges across all data collected in Trial 1, and at each passenger count location (within bus)

| | Ground Truth ($n_{gt}$) | Estimated ($\hat{n}$) | Accuracy (%) | MAE | RMSE |
|---|---|---|---|---|---|
| Front In | 1539 | 1502 | 97.6 | 1.11 | 2.14 |
| Front Out | 412 | 460 | 88.4 | 0.77 | 1.54 |
| Rear In | 39 | 318 | 12.26 | 2.74 | 10.01 |
| Rear Out | 1136 | 1126 | 99.92 | 5.97 | 5.30 |
| **Total** | **3126** | **3406** | **91.8** | **2.65** | **4.75** |

**Table 5**
Off-bus passenger count accuracy as percentage of total passenger exchanges across all data collected in Trial 2, and at each location (passenger pick-up sites)

| | Ground Truth ($n_{gt}$) | Estimated ($\hat{n}$) | Accuracy (%) | MAE | RMSE |
|---|---|---|---|---|---|
| Reservoir | 4082 | 4016 | 98.4 | 7.66 | 12.13 |
| Parliament | 4583 | 4532 | 98.9 | 5.03 | 7.58 |
| Pakenham | 657 | 479 | 72.9 | 4.16 | 6.77 |
| Dandenong | 1470 | 1401 | 95.3 | 2.80 | 4.17 |
| **Total** | **10,792** | **10,428** | **96.6** | **4.91** | **7.66** |

**Table 6**
Quartile analysis of net passenger exhange estimate accuracies obtained using the video-based APC system. Table reports distributions for combined entry/exit points (Front Door ∪ Rear Door In/Out), each doorway in isolation (Front Door In/Out and Rear Door In/Out), and entry and exit events in isolation at each doorway (Front In, Front Out, Rear In, Rear Out)

| | Bus | Front Door | Rear Door | Front In | Front Out | Rear In | Rear Out |
|---|---|---|---|---|---|---|---|
| Min | 0% | 0% | 0% | 0% | 0% | 0% | 0% |
| Q1 | 36% | 67% | 2% | 86% | 100% | 100% | 21% |
| Median | 75% | 100% | 100% | 100% | 100% | 100% | 100% |
| Q3 | 100% | 100% | 100% | 100% | 100% | 100% | 100% |
| Mean | 78% | 83% | 73% | 84% | 84% | 92% | 69% |

### 4.5. Quartile accuracies

A quartile distribution analysis was conducted to inspect the distribution of accuracies achieved for *per-bus-stop* passenger exchanges (on-bus) and *per-bus* passenger boarding counts (off-bus) across both trials. This was undertaken to address the primary use-case for both systems, maintaining on-going, real-time estimates of bus occupancy.

Table 6 and Fig. 8 present quartile accuracy distributions and provide associated box-plots respectively for the on-bus video-based APC system's per-stop net passenger exchange performance. That is, the accuracy of the method to determine the number of passengers that entered the bus minus the number that exited the bus at each stop, across all data collected from both buses. Quartile accuracies thus represent the distribution of net passenger exchange accuracies achieved over all bus stops during Trial 1.

**Table 7**
Quartile distribution of per-bus passenger boarding counts obtained using the video-based APC system in Trial 2 (off-bus). Table reports distributions for all passenger pickup locations (All), as well as for each location in isolation. Where two cameras were used (Camera 1 and 2 at Reservoir and Parliament), quartile accuracies are reported for each camera seperately. Note that only Camera 1 data was used in the quartile distribution analysis across all sites.

| | All | Reservoir | | Parliament | | Pakenham | Dandenong |
|---|---|---|---|---|---|---|---|
| | Cam 1 | Cam 1 | Cam 2 | Cam 1 | Cam 2 | Cam 1 | Cam 1 |
| Min | 20% | 24.7% | 0.0% | 11.9% | 2.4% | 0.0% | 26.7% |
| Q1 | 79.6% | 79.5% | 70.8% | 82.2% | 58.4% | 55.8% | 83.3% |
| Median | 88.9% | 88.9% | 82.1% | 90.6% | 74.1% | 80.0% | 90.5% |
| Q3 | 97.0% | 97.8% | 94.8% | 96.8% | 90.8% | 91.9% | 97.1% |
| Mean | 84.0% | 84.4% | 76.6% | 85.2% | 74.3% | 69.1% | 85.1% |

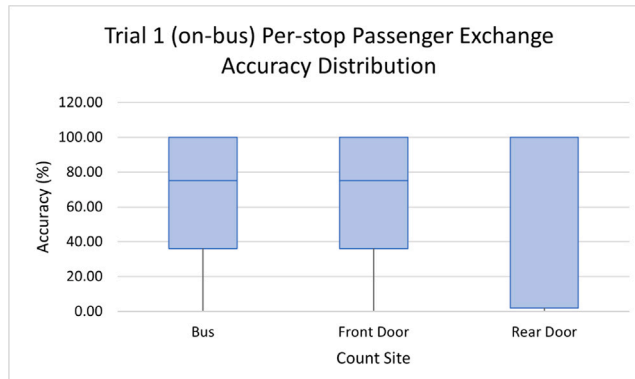


**Fig. 8.** Box plots from Trial 1 (on-bus) data showing distribution of accuracies achieves for per-stop passenger exchanges as estimated by the video-based APC system. Plots are given for entire bus passenger exchanges (i.e., through both doors) and when considering each bus door in isolation. Quartile accuracies corresponding to these plots are given in Table 6.

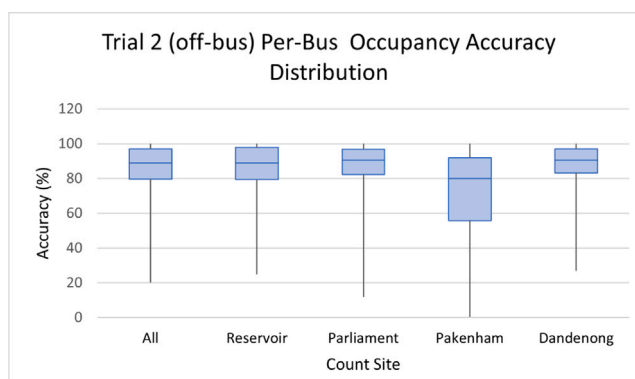


**Fig. 9.** Box plots from Trial 2 (off-bus) data showing distribution of accuracies achieves for *per-bus* passenger boarding counts as estimated by the off-bus video-based APC system (Camera 1). Plots are given for entire all passenger pick-up sites across the trials, as well as each site in isolation. Quartile accuracies corresponding to these plots are given in Table 7.

Similarly, Table 7 and Fig. 9 report quartile accuracy distributions (values and box-plots respectively) for the Trial 2 (off-bus) per-bus occupancy estimation use-case. In this case, quartile accuracies indicate the distribution of bus occupancy estimation accuracies over all buses that arrived at a passenger pick-up location during Trial 2.

The left most column of reported accuracy distributions in Tables 6 and 7 report the median accuracy achieved for total net passenger exchanges in both Trial 1 (on-bus) and Trial 2 (off-bus) scenarios.

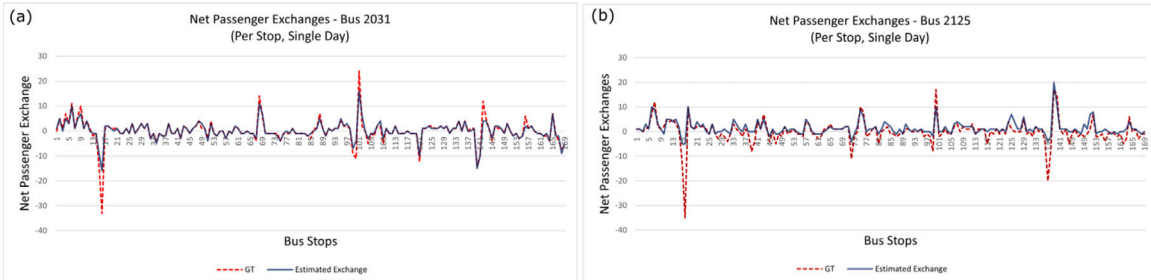


**Fig. 10.** Sample Trial 1 (on-bus) plots comparing estimated net passenger exchanges across a single day of the trial. Estimated net passenger exchanges are calculated from the combined front and rear door passenger enter/exit count estimates. Plots are given for data collected from Bus 2031 on (a) Day 1 and (b) Day 2.

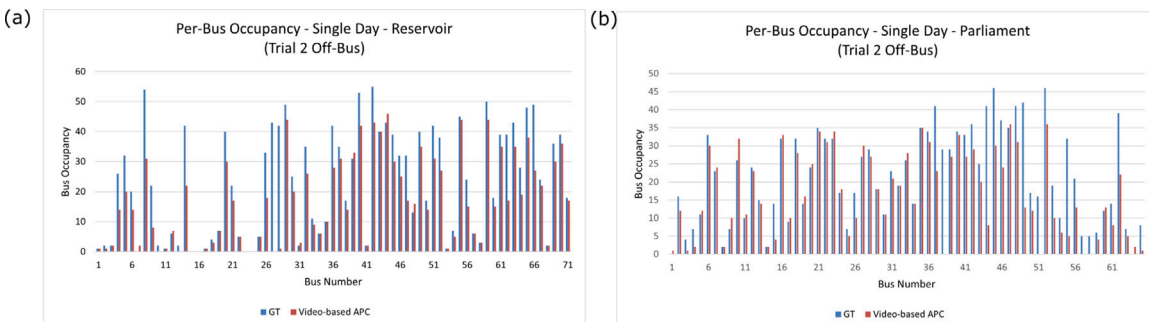


**Fig. 11.** Sample Trial 1 (off-bus) plots comparing estimated passenger boarding counts with ground truth across two single days of the trial, a locations (a) Reservoir, and (b) Parliament.

The off-bus video-based APC achieves the highest median accuracy of 88.9% (and mean 84%), compared with just 75% (mean: 78%) for the on-bus APC. Box plots for both trials (Figs. 8 and 9) show that the off-bus scenario also exhibits a substantially narrower distribution of accuracies, suggesting an overall more stable performance compared with the on-bus use-case.

There are a number of use-case specific factors that are likely contributing to this observed difference in performance. Most prominent is the off-bus system's exclusive focus on boarding passengers only through a single door. In contrast, the on-bus system derives net passenger exchange estimates from the combined count of two cameras within the bus, and covers passenger flow in both directions. It thus represents a more complex networked system with more sources of error. Indeed, Fig. 8 suggests the relative drop in accuracy of the on-bus net passenger estimates compared with off-bus is primarily due to a very wide spread of accuracies achieved using the rear-door APC system. Considering front-door entries only however, Table 6 shows the on-bus system achieves comparable accuracy to the off-bus system (median: 100%, mean: 84%), including a higher median accuracy compared with off-bus.

Figs. 10 and 11 show net passenger exchanges across entire sample days for Trial 1 and Trial 2 respectively, providing further insight into differences between the two scenarios. Specifically, it is evident that a higher frequency of passenger exchange events was experienced in the on-bus setting, however, net passenger exchanges per stop generally limited to less than ten. In contrast, Trial 2's *en-masse* passenger pick-up events were less frequent, but more often than not exceeded 10 passengers. This difference in passenger count patterns is well explained by the nature of the rail replacement services upon which counting was performed in Trial 2, compared with the regular weekly bus services from which Trial 1 data was predominantly captured from. It does, however, highlight implications for the accuracy of bus occupancy estimates obtained using the video-based APC. Closer inspection of the plots in both Figs. 10 and 11 shows that the video-based APC system appears to qualitatively capture the pattern of passenger exchanges with reasonable accuracy in both trial settings. Errors, in general, appear to be largest for net passenger exchange events involving larger numbers of passengers. While evident in both scenarios, it is more apparent in the on-bus setting, suggesting occlusions in the more cluttered setting are impacting the count - in particular for front-door exits. Fig. 12 illustrates this issue with selected examples where passenger occlusion is in play. Occlusions in the off-bus setting were largely avoided by virtue of the camera being positioned at right angles to the flow of passengers.

## 5. Discussion

The presented field trial results compare the performance of a state-of-the-art video-based automatic people counting system for the task of passenger counting in the rail replacement bus context. While specific to the public transport context, we argue this use-case provides an ideal focus point for better understanding the key considerations and implications of video-based solutions for crowd counting and monitoring more generally. To this end, below we discuss the broader technical, operational and planning implications of these results both for passenger counting, and video-based crowd management applications.

### 5.1. Counting accuracy implications

The net passenger count accuracies reported for Trial 1 (on-bus) and Trial 2 (off-bus) were 91.8% and 96.6% respectively, results comparable with reported accuracies for similar solutions targeting low-powered, edge-based solutions using video such as (Perng et al., 2016; Haq et al., 2020; Ghaziamin et al., 2024). This result supports the underlying assertion that the method deployed, while not the most advanced method available, was representative of state-of-the-art solutions suitable for the scenarios presented.

Net passenger count accuracies from both trials suggest the off-bus scenario yielded more accurate counts. However, it is interesting to note that larger differences in performance occurred for both on-bus and off-bus scenarios when comparing accuracies at different points of capture and/or times of day. In the off-bus setting, for example, the

**Fig. 12.** Sample images from front on-bus camera showing exhibiting (a) alighting passengers, and (b) boarding passengers. In general, boarding passengers were detected earlier and tracked for longer prior to crossing the count line compared with alighting passengers.

lowest net count accuracy was achieved at Pakenham (mean accuracy of 72. 1%) compared to the best performing location, the reservoir (mean accuracy of 98.27%). This difference has been firmly attributed to the poor lighting conditions at Pakenham (discussed later). On-bus results reflected an even greater range of accuracies with respect to the location of the camera, with the front camera achieving a mean accuracy of 93% versus the rear door's mean accuracy of 56.09%. In this case, poor visibility and less predictable passenger movement in and around the rear door are likely causes. In general, these accuracy variations reflect the impact of very specific environmental factors such as lighting, passenger movement patterns, and visual coverage of the space that have the most pronounced influence on performance.

### 5.2. Installation and configuration implications

#### 5.2.1. Position of camera

The impact of camera positioning was apparent across both trials. The off-bus positioning of the camera close to the bus, aligned side-on with the flow of passengers achieved optimal results. The alternate off-bus camera position, at 45 degrees to the passenger flow also supported strong accuracy results, suggesting any view that maximises coverage of passenger movement as they approach the count line, while also minimising the possibility of occlusions is sufficient for video-based APC. In the on-bus context this is further supported by results for the front-door boarding passenger counts. While moderately less accurate compared with off-bus camera positions, this view maximised coverage of the passenger flow and minimised occlusions within the constrained bus setting.

The inherently limited choice of camera positions within the confined space of the bus was an impeding factor for the on-bus system's accuracy, compared to the off-bus scenario. The rear door exemplified this issue, with no camera position adequately able to provide a clear un-occluded line of sight of alighting passengers. While the front door camera was better positioned, occlusions evidently impacted the count accuracy for this door as well — in particular front-door exits.

Alternative on-bus camera positions may improve on the results presented here — most obvious being above the door-way, top-down. Preliminary investigations of this position were undertaken in Trial 1, however the video-based APC method applied was necessarily modified to handle the top-down perspective, and utilised a depth-sensing (RGB+Depth) camera. In practise, however, and in particular with respect to the temporary rail replacement bus scenario, above the door-way also presents potential challenges for the secure and stable installation of the device compared with horizontal mount positions, which may be more readily installed on vertical bus hand rails.

#### 5.2.2. Calibration

System calibration is an essential and unavoidable installation step for video-based line-of-interest counters. In both field trials, the ability to calibrate on-site prior to counting was essential to the system's overall level of accuracy, but also the system's flexibility in supporting a wide range of different camera positions and passenger flow directions with respect to the camera's viewing angle. Calibration was observed to be particularly critical for on-bus installation, where camera positions were limited due to constraints on physical camera mount points within the bus; this included device security concerns. In the off-bus setting, calibration was also key to the system's ability to be configured for entirely different camera positions. Any future deployed system would need to ensure that an easy user-friendly human-machine interface is available to support ground staff in performing such calibration.

### 5.3. Operational implications

From an operational perspective, both trials highlighted benefits and challenges specific to each use-case. The on-bus scenario, as an indoor setting, benefited from the relatively stable visual conditions in which it operated. Sufficient lighting was always present, though subject to variation during bus travel, and the relative stability of the system's accuracy across all trial days in general suggests that the system was unaffected by variable weather conditions or other external factors. In contrast, the off-bus system did experience several challenges in this regard, reflected in more variation of accuracies across the trial compared with the on-bus scenario. Most notable was the impact of lighting extremes - for example, the pre-sunrise mornings at Pakenham, but also observed periods in the later afternoon (at Dandenong) when light saturation due to direct sunlight was observed. Also of note was the impact of heavy rain, which was observed to decrease image capture quality but also alter passenger behaviour during bus boarding. Detection accuracy was evidently impeded by increased occlusions due to the use of umbrellas and other implements to shelter from rain, as well as less orderly bus entry behaviour in general. Fig. 13 shows examples of this.

From a device monitoring perspective, the off-bus system was easiest to manage and monitor. While both systems were supervised at all times by research team members, the off-bus system's fixed geographical location during data capture supported reliable remote connectivity with the device, and thus easy local monitoring. Any issues that arose were quickly dealt with on site. For the on-bus system, while continuous connectivity was theoretically possible, this could not be safely assumed during the trial due to variable cellular network coverage of areas serviced by the bus. Moreover, any issues that arose invariably required direct access to the hardware, and only when the bus reached a depot or long duration stop such that the device could be accessed safely and unhindered.

In general, the trials made clear that while off-bus deployment of the system presented more external variables to account for, on-bus deployment placed greater demand on ensuring the system's reliability and correct configuration. This is important to consider in the context of temporary use-cases such as for rail replacement bus services, where

**Fig. 13.** Example images from off-bus system during heavy rain. Occlusions due to umbrellas were an observed source of error for the video-based APC under these conditions.

bus access prior to use is limited, and buses servicing the route may vary significantly in design and interior layout. In contrast, off-bus passenger pick-up and put-down locations are generally known in advance (eg., in Melbourne, most railway stations have a dedicated bus zone) and thus may be planned for in advance, and in general offer more flexibility for camera positioning and configuration.

### 5.4. Technology implications

The video-based APC system deployed in both trials is representative of a state-of-the-art detection-based approach. However, the system could be improved to optimise performance based on the outcomes of these trials to address observed shortcomings in both scenarios. Below lists key areas where improvements could be made, based on the observed performance of the system

#### 5.4.1. Accuracy

The trialled system deployed a pre-trained Convolutional Neural Network (MobileNetv3), trained to detect 90 different generic object classes using crowd-sourced annotated image data. One of these classes is people, however further data collection and more refined use-case specific training would improve the accuracy of detection, particularly for exiting passengers in the on-bus setting. Increased robustness to other scenarios such as umbrellas and other common causes of occlusions could also be supported through more directly targeted training.

Modelling systematic error has also been explored to account for statistically predictable errors in occupancy estimation. In both scenarios, a simple linear regression model may be trained to account for such error. In a preliminary investigation of this reported in McCarthy et al. (2021), it is shown that approximately 94% of the observed accuracy error in on-bus passenger exchange estimates can be accounted for by such a model. This suggests accurate ongoing bus occupancy estimates should be achievable despite the presence of detection and tracking errors.

#### 5.4.2. Image capture

The ability to support different field of view configurations was observed as a desirable but missing feature of the system in both use-cases. The system deployed an RPi Camera Module with fixed focus lens, offering a 62° horizontal field of view. Alternative lens choices may reasonably be expected to improve the performance of the system, particularly in the more confined space of the on-bus scenario — especially coverage of the rear-door. In the off-bus scenario, while less of concern, a wider field of view could be considered to support associated use-cases such as monitoring passenger wait time, or automatically detecting bus arrivals (and Bus IDs). Lens filter choices may also be incorporated to handle different lighting extremes, including direct sunlight.

#### 5.4.3. Device

The housing of the device for both scenarios was considered carefully, however could be improved. Off-bus presented the most challenges due to its exposure to the elements. Wind, rain and sun all presented challenges to the physical device's stability and scene visibility in the images captured. While the system was designed to be waterproof, it was observed on the day that the video capture system should also incorporate some form of visor to shelter the lens from direct rain fall, which was seen to degrade the visibility of the scene.

It should be noted that while the above improvements may reasonably be expected to improve the video-based APC system's performance, the observed challenges and trade-offs associated with deploying the system in the on-bus and off-bus scenarios are unlikely to be fully addressed by any single near-term advancement. It is thus important to understand the working limits of video-based solutions such as the one trialled here to make appropriate use-case and configuration choices.

### 5.5. Line-of-interest counting versus head-counts

The trialled video-based APC system performs line-of-interest counting, necessitating its use as an estimator of net passenger exchanges at the point of passenger entry/exit in both trial settings. An alternative configuration for video-based counting is to obtain occupancy estimates via direct passenger head-counts within single frame views of the scene (i.e, more generally referred to as *region-of-interest* methods, as discussed in Section 2.2) (Labit-Bonis et al., 2021). The key advantage of such an alternative is the removal of accumulating error associated with aggregating net passenger exchanges over time (e.g., per bus stop), however this requires sufficient visual coverage and line of sight with all passengers within a single view (or collection of views) to obtain counts reliably. In the on-bus setting this may be plausible but is made challenging by the narrow and relatively cluttered environment presented, thereby increasing the likelihood of occlusions — especially when the bus is near full capacity. While this is also true of detection-based methods, as observed in the presented results and in previous work (e.g., Hsu et al., 2020b), the more localised viewing area required for line-of-interest counting reduces the impact of this issue. In the off-bus setting, the application of head-counting approaches is made complicated due to the challenge of correctly associating head-counts with those people actually boarding or alighting from the bus. This would likely rely on additional infrastructure to guide passengers to queuing areas. However, given this infrastructure was in place for Trial 2, a line-of-interest counter would seem the better suited configuration when passenger flow is constrained to a predominantly linear path.

Line-of-interest counters arguably represent the simpler system to install and configure by virtue of the more constrained visual area they must cover. Head-counting methods, conversely, must either utilise pre-existing CCTV cameras (which may or may not offer sufficient image quality or coverage sufficient for counting), or be installed for purpose and thus with more care to ensure an optimal view of the counting area. In the temporary, rail replacement setting, this is likely to add substantial time to the installation and setup, and for minimal accuracy benefit given other challenges as noted above.

### 5.6. Video-based APC versus alternative technology choices

Interest in automatic people counting is motivated by a wide range of crowd management and monitoring needs, spanning real-time use-cases such as demand monitoring such as in the public transport context explored here, to data capture for future crowd management planning. The suitability of APC technologies thus depends critically on the underlying needs, and will vary based on differing priorities. Previous work in the passenger counting and monitoring context has explored a multitude of sensors and technologies to support APC, highlighting a range of trade-offs with respect to accuracy, power efficiency, cost, robustness, flexibility, privacy and device/data security among other use-case specific factors (Kouyoumdjieva et al., 2019). The presented results and observations here support the view that while video-based APC offers generally high accuracy and ability to support a wide range of use-cases, it is also the most susceptible technology to external variables, such as weather, lighting and crowd behaviour, thus highlighting its reliance on appropriate environmental management. In contrast, other APC technologies, such as radio, LED, Passive Infrared, or Pressure/Force sensing in general offer a very high degree of counting accuracy, and with minimal susceptibility to environmental factors. Thus, where continuous very high counting accuracy is sought, then such options may be the more suitable choice. However, these technologies offer only singular modalities of operation, and in most cases, impose other working limits such as a high reliance on passenger flow conforming to the needs of the sensors. Other APC approaches such as Wi-Fi polling methods alleviate this issue, but present their own set of challenges with respect to polling time requirements and the issue of limiting head counts to confined regions of interest (e.g., within a bus, or a bus boarding zone).

The results presented support the view that video-based APC is most suitable when accuracy is of moderate to high importance, but not mission critical to the decision making outcomes at a given time during data capture, when flexibility of installation sites is desirable, and timely device installation and removal is a consideration. Video-based APC is arguably best placed of all APC technologies to support additional data capture use-cases such as analysing crowd wait/dwell times, assessing people movement more generally, automatically detecting Vehicle ID/registration signage, amongst a multitude of additional information present in the visual image data. The ability to extract such information at low cost, at scale, and with minimal maintenance makes video-based people monitoring an attractive option for such scenarios. Furthermore, Video-based APCs are able to capture other environmental and object related information, when needed (e.g. gender, disability, accompanying animals, etc.)

### 5.7. Video-based APC selection implications

In this paper we present the results of trialling a video-based APC solution, along with two set-up configurations of on- and off-vehicle. The accuracy results and implications related to the installation and environmental factors provide important insights for managers when selecting the appropriate video-based APC approach for their desired needs. In the case of rail replacement buses, the service is predominantly characterised as being temporary in nature with high spatio-temporal volatility. For such services, the preliminary use of APC is to establish a better understanding of supply (available buses) and demand (patronage) between main points of passenger flow (e.g., start and end transit station on the disruption line). For such applications, off-bus video-based APCs appear to be a the appropriate configuration due to lower number of required data collection points and ease of installation. Furthermore, the accuracy reported by video-based APCs is ample to understand passenger demand at points of interest. Another operational factor influencing the choice of on- and off-bus solution is associated with the ownership of assets, specially vehicles. Currently, a large portion of bus services is provided by the private sector, mainly large companies with operations across the globe. Therefore, access, installation and data capturing from vehicles by third parties (e.g. researchers and government) come with challenges when such arrangements are not contractually established. Hence, off-bus solution appears to be operationally more appropriate. On-bus video-based APCs, on the other hand, provide a more long-term and permanent solution for bus operators to better understand vehicle utilisation and patronage patterns more methodically and consistently. However, most vehicles are not equipped with video-based APC when manufactured, requiring significant retrofitting, installation and integration.

## 6. Conclusion

Video-based automatic people detection and monitoring is emerging as an IoT solution of interest for a range of crowd management and monitoring applications. Public transport is one such application, for which rail replacement bus services present a particularly compelling use-case for APC technologies as a means of improving the accuracy and completeness of data capture for real-time feedback and monitoring, demand forecasting and future event/site planning. Video-based APC represents a particularly versatile and flexible solution, but comes with important constraints with regards to how it is deployed and used, especially with respect to variable environmental conditions and external factors such as lighting, weather, and occlusions. In the context of rail replacement buses, a fundamental choice is whether to deploy such systems on-bus, or off-bus.

In this paper we have presented results from two independently conducted field trials of a state-of-the-art video-based APC method in which the system was deployed for both on and off bus passenger counting use-cases. The results presented indicate that the video-based approach offers good levels of accuracy overall in both scenarios, though performs strongest when the camera is positioned side on to the dominant flow of passenger traffic, and with sufficient visual coverage to support robust tracking. The off-bus (ie., on-ground) scenario best supports this configuration, however was observed to be significantly more impacted by external factors such as weather and lighting. The on-bus scenario, in contrast, was observed to be largely immune to such factors, however was more susceptible to occlusions due to high passenger density within the confined space of the bus. Visual coverage of doorways was also noticed as a challenge, as well as avoiding false positive counts when the bus approaches full capacity (i.e., passengers standing in door-ways). Other differences between the two deployment scenarios were also observed, with our analysis suggesting the off-bus deployment scenario offered, from an installation and operational standpoint, the least complicated and most scalable use-case for video-based APC. More significantly, however, our results make clear the need for extensive field trialling when determining which APC technology to use. While video-based APC offers some clear benefits over alternative technology choices, its performance relies critically on its configuration and calibration, and for what purpose it is deployed.

It is important to note that this research has focused exclusively on detection-based line-of-interest passenger counting, and a particular implementation of such an approach. Future research may consider other recent methods, as well as alternative video-based APC methods and observe different accuracy results — in particular methods that seek to acquire total head-counts rather than line-of-interest counting. Overall, however, the operation considerations associated with the use of standard RGB video data for passenger counting highlighted are inherent to vision-based solutions generally, and while may be mitigated through algorithmic and design advancements, are likely to remain important considerations regardless.

## CRediT authorship contribution statement

**Chris McCarthy:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Hadi Ghaderi:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Prem Jayaraman:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Hussein Dia:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Chris Mccarthy reports financial support was provided by iMOVE CRC (Australia). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

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